CEG 4750/6750 Information Security

Lab/Project 2

Using AES Algorithm

A project submitted in partial fulfillment of the requirements for the

CEG 4750/6750 Information Security

By

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To

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Introduction

In this project, the AES encryption/decryption programs are provided, and the students (we) will compile and run these programs to encrypt/decrypt files on *fry.cs.wright.edu*. The AES encryption/decryption programs used an existing crypto library, Crypto++ (C++), which was installed on *fry.cs.wright.edu*.

For the project, the very first thing we did was to connect to *fry.cs.wright.edu* and access the shell to do the work. To keep things simple, we created directory named *Project 2*, copied files into the folder and began working under the directory.

As per the instruction provided and for our own ease, we created an alias called *cryptog++* for the command as shown in the Figure 1.1 below. This helped us avoiding typing such a long command each time we try to compile our program.

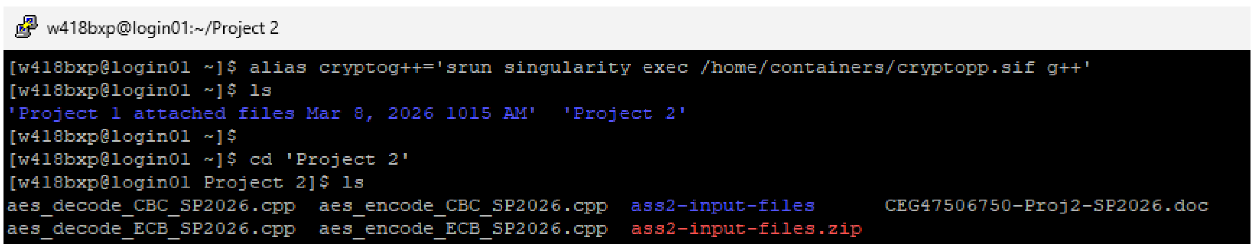

Figure 1.1 – Alias

Also as demonstrated in the below Figure 1.2, we can see that the alias is working perfectly as the compilation of both the encryption/decryption program works. The encryption and decryption programs are named *Task1CbcEnc* and *Task1CbcDec* respectively.

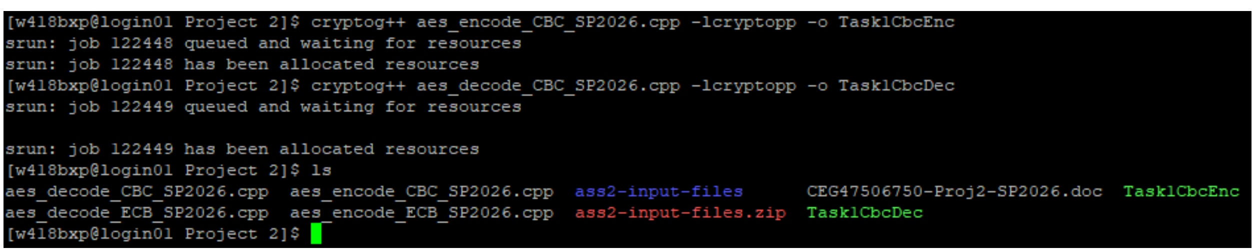

Figure 1.2 – Compilation

Now, we begin along with the task that we are supposed to complete.

Task 1

For Task 1 and Task 2, we are required to compile and execute the provided source AES encryption/decryption programs in CBC and Counter mode respectively, and verify the decrypted file is identical to the original plaintext file. There are total of 4 test files provided to us for the task, which are *MSG1*, *MSG2*, *MSG3*, and *text1*. And for each file, we are required to do the following:

1. Encrypt the file to output it suffixed with *.e* to indicate it as an encrypted file.
2. Decrypt the file to output it suffixed with *.d* to indicate it as a decrypted file.

3. Compare the bytes of couple of encrypted files.
4. Show the difference between two files (the original and the decrypted) to demonstrate the working nature of overall process.

For the subtasks requested, we began encrypting each of the input files one by one as shown in the Figure 1.3 below.


```
w418bxp@login01:~/Project 2
[w418bxp@login01 ~]$ cd 'Project 2'
[w418bxp@login01 Project 2]$ ls
aes_decode_CBC_SP2026.cpp  aes_encode_CBC_SP2026.cpp  ass2-input-files      CEG47506750-Proj2-SP2026.doc  MSG1.e  MSG3         Task1CbcEnc
aes_decode_ECB_SP2026.cpp  aes_encode_ECB_SP2026.cpp  ass2-input-files.zip  MSG1                          MSG2    Task1CbcDec  text1
[w418bxp@login01 Project 2]$ ./Task1CbcEnc MSG1 MSG1.e sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
plain text: The most serious incident in NASA since 1986.

cipher text stored in:MSG1.e
[w418bxp@login01 Project 2]$ ./Task1CbcEnc MSG2 MSG2.e sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
plain text: The most serious accident in NASA since 1986.

cipher text stored in:MSG2.e
[w418bxp@login01 Project 2]$ ./Task1CbcEnc MSG3 MSG3.e sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
plain text: The most serious incident in NASA since 1985.

cipher text stored in:MSG3.e
[w418bxp@login01 Project 2]$ ./Task1CbcEnc text1 text1.e sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
plain text: The loss of the space shuttle Columbia with seven astronauts on board
on Saturday was the most serious incident in NASA's shuttle program
since the 1986 explosion of the Challenger, in which seven astronauts died.

The following are landmark missions in the NASA Space Shuttle program.

1981, Apr 12-14 -- The space shuttle Columbia is launched from
Kennedy Space Center, Florida, to begin the first shuttle mission. I
ts primary objectives were to accomplish a safe ascent into orbit,
check all flight systems and carry out a safe landing.
After landing it was found that a take-off pressure wave had
displaced 16 heat shield tiles and damaged 148 others.

1981, Nov 12 -- Columbia undertook the second shuttle mission,
shortened from five to three days due to fuel cell failure,
after several launch delays. NASA said 90 percent of mission
objectives had been achieved.

1982-85 -- Shuttles make 20 missions into orbit and back.

1983 June 18-24 -- First U.S. woman astronaut, Sally Ride,
takes part in NASA's seventh shuttle mission, on board Challenger.

1986, Jan 28 -- The shuttle Challenger explodes 72 seconds after
take-off, killing the seven crew, including the schoolteacher
Sharon C. McAuliffe. Blast caused by a leak of liquid hydrogen.
Shuttle flights were halted while extensive investigation into accident
and assessment of Shuttle program were conducted.

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ly flights and crew exchange flights continue through 1999, 2000 and 2001.

2002 Nov-Dec -- Endeavour exchanges the ISS's fifth and sixth mission crews in a 14-day mission in orbit. It marks Endeavour's 19th flight.

2003 Feb 1 -- Columbia, the 113th shuttle mission, carrying Ilan Ramon, the first Israeli in space, breaks up over Texas with seven astronauts on board after
 16 days in orbit.


cipher text stored in:text1.e
[w418bxp@login01 Project 2]$
```

Figure 1.3 – Encryption of input files

Similarly, we tried decrypting the encrypted files as shown in the following Figure 1.4 and Figure 1.5.

w418bxp@login01:~/Project 2

```
[w418bxp@login01 Project 2]$ ./Task1CbcDec MSG1.e MSG1.d sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
recovered text: The most serious incident in NASA since 1986.

plain text stored in:MSG1.d
[w418bxp@login01 Project 2]$ ./Task1CbcDec MSG2.e MSG2.d sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
recovered text: The most serious accident in NASA since 1986.

plain text stored in:MSG2.d
[w418bxp@login01 Project 2]$ ./Task1CbcDec MSG3.e MSG3.d sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
recovered text: The most serious incident in NASA since 1985.

plain text stored in:MSG3.d
[w418bxp@login01 Project 2]$ ls
aes_decode_CBC_SP2026.cpp  aes_encode_ECB_SP2026.cpp  CEG47506750-Proj2-SP2026.doc  MSG1.e  MSG2.e  MSG3.e       text1
aes_decode_ECB_SP2026.cpp  ass2-input-files           MSG1                          MSG2    MSG3    Task1CbcDec  text1.d
aes_encode_CBC_SP2026.cpp  ass2-input-files.zip       MSG1.d                        MSG2.d  MSG3.d  Task1CbcEnc  text1.e
[w418bxp@login01 Project 2]$
```

Figure 1.4 – Decryption of encrypted files

```
[w418bxp@login01 Project 2]$ ./Task1CbcDec text1.e text1.d sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
recovered text: The loss of the space shuttle Columbia with seven astronauts on board
on Saturday was the most serious incident in NASA's shuttle program
since the 1986 explosion of the Challenger, in which seven astronauts died.

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 16 days in orbit.


plain text stored in:text1.d
[w418bxp@login01 Project 2]$ ls
aes_decode_CBC_SP2026.cpp  aes_encode_ECB_SP2026.cpp  CEG47506750-Proj2-SP2026.doc  MSG1.e  MSG2.e  MSG3.e       text1
aes_decode_ECB_SP2026.cpp  ass2-input-files           MSG1                          MSG2    MSG3    Task1CbcDec  text1.d
aes_encode_CBC_SP2026.cpp  ass2-input-files.zip       MSG1.d                        MSG2.d  MSG3.d  Task1CbcEnc  text1.e
[w418bxp@login01 Project 2]$
```

Figure 1.5 – Decryption of encrypted files

Task 1.1

So far, so good. For the subtask 1.1, we print the octal byte values using the command *od –b filename* for each file as shown in the below Figure 1.1.1.

```
[w418bxp@login01 Project 2]$ od -b MSG1.e
0000000 344 011 306 332 043 251 274 252 373 337 006 004 020 201 120 115
0000020 373 307 325 105 174 255 074 070 024 311 267 071 044 333 115 362
0000040 072 003 254 123 361 343 033 120 222 321 037 216 337 353 376 251
0000060
[w418bxp@login01 Project 2]$ od -b MSG2.e
0000000 344 011 306 332 043 251 274 252 373 337 006 004 020 201 120 115
0000020 215 312 127 007 322 354 337 352 353 152 166 326 170 060 230 123
0000040 115 331 247 261 162 050 254 067 220 121 011 205 133 241 336 213
0000060
[w418bxp@login01 Project 2]$ od -b MSG3.e
0000000 344 011 306 332 043 251 274 252 373 337 006 004 020 201 120 115
0000020 373 307 325 105 174 255 074 070 024 311 267 071 044 333 115 362
0000040 050 363 304 374 354 351 307 063 140 233 343 176 130 032 227 141
0000060
[w418bxp@login01 Project 2]$
```

Figure 1.1.1 – Octal byte values of encrypted files

Task 1.2

From the above Figure 1.1.1, we can conclude for the comparison of MSG1.e and MSG2.e, the first 16 bytes (first two rows) are completely identical. The difference begins at byte 16 (offset 0000020) and continues through the remaining bytes. This indicates that the first AES block (16 bytes) is identical in both ciphertexts, while subsequent blocks differ.

Similarly, for MSG1.e and MSG3.e, the first two rows (first 32 bytes) are identical and the difference only starts at byte 32 (offset 0000040), meaning the first two AES blocks (32 bytes) are identical, with differences beginning at the third block.

This overall difference is actually expected with AES-CBC mode as messages share common prefixes and is 16 of bytes.

Task 1.3

We are required to use *diff* command to show the difference of two files (the original file, and the decrypted file). The result is shown in the Figure 1.3.1 below:

w418bxp@login01:~/Project 2

```
[w418bxp@login01 Project 2]$ diff MSG1 MSG1.d
[w418bxp@login01 Project 2]$ diff MSG2 MSG2.d
[w418bxp@login01 Project 2]$ diff MSG3 MSG3.d
[w418bxp@login01 Project 2]$ diff text1 text1.d
[w418bxp@login01 Project 2]$
```

Figure 1.3.1 – Difference between files

We can observe from the above Figure 1.3.1, we can clearly conclude that the encryption and decryption works perfectly as there are no differences shown for all the comparisons made.

As we have noticed already, the CBC mode requires an initialization vector (IV) to encrypt and decrypt a file where the size of the IV is set to the AES's default length. i.e. 16 bytes typically. The source code for both the files are provided in below Figure 1.6 and Figure 1.7.

```cpp
// CEG 47506750 sample code

#include<iostream>
#include<fstream>
#include<sstream>
#include<string>
using namespace std;

#include"cryptopp/cryptlib.h"
#include"cryptopp/hex.h"
#include"cryptopp/filters.h"
#include"cryptopp/des.h"
#include"cryptopp/aes.h"
#include"cryptopp/modes.h"

using namespace CryptoPP;

string aes_encode(string & plain,byte key[], byte iv[])
{
    string cipher;
    try{
        CBC_Mode<AES>::Encryption enc;
        enc.SetKeyWithIV(key, AES::DEFAULT_KEYLENGTH, iv);
        StringSource(plain, true, new StreamTransformationFilter(enc, new StringSink(cipher)));
    }
    catch(const CryptoPP::Exception & e)
    {
    }
    return cipher;
}

int main(int argc, char * argv[])
{
    fstream file1;
    fstream file2;
    byte key[AES::DEFAULT_KEYLENGTH];
    byte iv[AES::BLOCKSIZE];

    if(argc!=4)
    {
        cout<<"usage:aes_encode infile outfile key"<<endl;
        return 0;
    }
    file1.open(argv[1],ios::in);
    file2.open(argv[2],ios::out);
    //reading
    stringstream buffer;
    buffer << file1.rdbuf();
    string plain(buffer.str());
    //cout<<"plain text:"<<plain<<endl;
    //get key
    memset(key,0,AES::DEFAULT_KEYLENGTH);
    for(int i=0;i<AES::DEFAULT_KEYLENGTH;i++)
    {
        if(argv[3][i]!='\0')
        {
            key[i]=(byte)argv[3][i];
        }
        else
        {
            break;
        }
    }
    memset(iv, 0, AES::BLOCKSIZE);
    //print key
    string encoded;
    encoded.clear();
    StringSource(key, sizeof(key), true, new HexEncoder( new StringSink(encoded)));
    cout << "key: " << encoded<< endl;
    //encode
    cout << "plain text: " << plain << endl;
    string cipher=aes_encode(plain,key, iv);
    file2<<cipher;
    cout<<"cipher text stored in:"<<argv[2]<<endl;

    file1.close();
    file2.close();
}
```

Figure 1.6 – Source code (aes_encode_CBC_SP2026.cpp)

```cpp
// CEG 47506750 sample code

#include<iostream>
#include<fstream>
#include<sstream>
#include<string>
using namespace std;

#include"cryptopp/cryptlib.h"
#include"cryptopp/hex.h"
#include"cryptopp/filters.h"
#include"cryptopp/des.h"
#include"cryptopp/aes.h"
#include"cryptopp/modes.h"

using namespace CryptoPP;
string aes_decode(string & cipher,byte key[], byte iv[])
{
    string plain;
    try{
        CBC_Mode< AES >::Decryption dec;
        dec.SetKeyWithIV(key, AES::DEFAULT_KEYLENGTH, iv);
        StringSource s(cipher, true, new StreamTransformationFilter(dec, new StringSink(plain)));
    }
    catch(const CryptoPP::Exception& e){
    }
    return plain;
}
int main(int argc,char * argv[])
{
    fstream file1;
    fstream file2;
    byte key[AES::DEFAULT_KEYLENGTH];
    byte iv[AES::BLOCKSIZE];

    if(argc!=4)
    {
        cout<<"usage:aes_decode infile outfile key"<<endl;
        return 0;
    }
    file1.open(argv[1],ios::in);
    file2.open(argv[2],ios::out);
    //reading
    stringstream buffer;
    buffer << file1.rdbuf();
    string cipher(buffer.str());
    //get key
    memset(key,0,AES::DEFAULT_KEYLENGTH);
    for(int i=0;i<AES::DEFAULT_KEYLENGTH;i++)
    {
        if(argv[3][i]!='\0')
        {
            key[i]=(byte)argv[3][i];
        }
        else
        {
            break;
        }
    }
    memset(iv, 0, AES::BLOCKSIZE);
    //print key
    string encoded;
    encoded.clear();
    StringSource(key, sizeof(key), true, new HexEncoder( new StringSink(encoded)));
    cout << "key: " << encoded<< endl;
    //decode
    string plain=aes_decode(cipher,key, iv);
    cout << "recovered text: " << plain<< endl;
    file2<<plain;
    cout<<"plain text stored in:"<<argv[2]<<endl;
}
```

Figure 1.7 – Source code (aes_decode_CBC_SP2026.cpp)

Task 2

For this task, we are provided with the source AES encryption/decryption programs that performs in Counter mode. We are required to change the source code files of the AES programs, so that the programs are working in Counter mode while still verifying that the decrypted file is identical to its original plaintext file.

This task in Counter mode, requires us to do perform the same type of operations as in Task 1. Meaning, we are required to encrypt and decrypt the files (with .ee and .dd suffix for encryption and decryption respectively), show the byte difference between MSG1.ee and MSG2.ee, MSG1.ee and MSG3.ee, and check the integrity with *diff* command for the input files and their respective decrypted ones.

As shown in below Figure 2.1, the encryption and decryption programs are named *Task2CounterEnc* and *Task2CounterDec* respectively.

```
w418bxp@login01:~/Project 2
[w418bxp@login01 Project 2]$ cryptog++ aes_encode_COUNTER_SP2026.cpp -lcryptopp -o Task2CounterEnc
srun: job 122450 queued and waiting for resources
srun: job 122450 has been allocated resources
[w418bxp@login01 Project 2]$ cryptog++ aes_decode_COUNTER_SP2026.cpp -lcryptopp -o Task2CounterDec
srun: job 122451 queued and waiting for resources
srun: job 122451 has been allocated resources
[w418bxp@login01 Project 2]$ ls
aes_decode_CBC_SP2026.cpp      aes_encode_COUNTER_SP2026.cpp  CEG47506750-Proj2-SP2026.doc  MSG2    MSG3.d       Task2CounterDec  text1.e
aes_decode_COUNTER_SP2026.cpp  aes_encode_ECB_SP2026.cpp      MSG1                          MSG2.d  MSG3.e       Task2CounterEnc
aes_decode_ECB_SP2026.cpp      ass2-input-files               MSG1.d                        MSG2.e  Task1CbcDec  text1
aes_encode_CBC_SP2026.cpp      ass2-input-files.zip           MSG1.e                        MSG3    Task1CbcEnc  text1.d
[w418bxp@login01 Project 2]$
```

Figure 2.1 – Compilation

For the subtasks requested, we began encrypting each of the input files one by one as shown in the Figure 2.2 and Figure 2.3 below.

```
w418bxp@login01:~/Project 2
[w418bxp@login01 Project 2]$ ./Task2CounterEnc MSG1 MSG1.ee sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
plain text: The most serious incident in NASA since 1986.

cipher text stored in:MSG1.ee
[w418bxp@login01 Project 2]$ ./Task2CounterEnc MSG2 MSG2.ee sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
plain text: The most serious accident in NASA since 1986.

cipher text stored in:MSG2.ee
[w418bxp@login01 Project 2]$ ./Task2CounterEnc MSG3 MSG3.ee sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
plain text: The most serious incident in NASA since 1985.

cipher text stored in:MSG3.ee
```

Figure 2.2 – Encryption of input files

```
[w418bxp@login01 Project 2]$ ./Task2CounterEnc text1 text1.ee sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
plain text: The loss of the space shuttle Columbia with seven astronauts on board
on Saturday was the most serious incident in NASA's shuttle program
since the 1986 explosion of the Challenger, in which seven astronauts died.

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ts primary objectives were to accomplish a safe ascent into orbit,
check all flight systems and carry out a safe landing.
After landing it was found that a take-off pressure wave had
displaced 16 heat shield tiles and damaged 148 others.

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shortened from five to three days due to fuel cell failure,
after several launch delays. NASA said 90 percent of mission
objectives had been achieved.

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takes part in NASA's seventh shuttle mission, on board Challenger.

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take-off, killing the seven crew, including the schoolteacher
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2003 Feb 1 -- Columbia, the 113th shuttle mission, carrying Ilan Ramon, the first Israeli in space, breaks up over Texas with seven astronauts on board after
 16 days in orbit.


cipher text stored in:text1.ee
[w418bxp@login01 Project 2]$
```

Figure 2.3 – Encryption of input files

The below Figure 2.4 with *ls* command shows that the above operations were executed successfully.

```
[w418bxp@login01 Project 2]$ ls
aes_decode_CBC_SP2026.cpp      aes_encode_COUNTER_SP2026.cpp  CEG47506750-Proj2-SP2026.doc  MSG1.ee  MSG2.ee  MSG3.ee       Task2CounterEnc  text1.ee
aes_decode_COUNTER_SP2026.cpp  aes_encode_ECB_SP2026.cpp      MSG1                          MSG2     MSG3     Task1CbcDec     text1
aes_decode_ECB_SP2026.cpp      ass2-input-files               MSG1.d                        MSG2.d   MSG3.d   Task1CbcEnc     text1.d
aes_encode_CBC_SP2026.cpp      ass2-input-files.zip           MSG1.e                        MSG2.e   MSG3.e   Task2CounterDec text1.e
[w418bxp@login01 Project 2]$
```

Figure 2.4 – List of files

Similarly, we tried decrypting the encrypted files as shown in the following Figure 2.5 and Figure 2.6.

```
w418bxp@login01:~/Project 2
[w418bxp@login01 Project 2]$ ./Task2CounterDec MSG1.ee MSG1.dd sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
recovered text: The most serious incident in NASA since 1986.

plain text stored in:MSG1.dd
[w418bxp@login01 Project 2]$ ./Task2CounterDec MSG2.ee MSG2.dd sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
recovered text: The most serious accident in NASA since 1986.

plain text stored in:MSG2.dd
[w418bxp@login01 Project 2]$ ./Task2CounterDec MSG3.ee MSG3.dd sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
recovered text: The most serious incident in NASA since 1985.

plain text stored in:MSG3.dd
[w418bxp@login01 Project 2]$
```

Figure 2.5 – Decryption of encrypted files

```
[w418bxp@login01 Project 2]$ ./Task2CounterDec text1.ee text1.dd sixteenbytesofkeys
key: 7369787465656E62797465736F666B65
recovered text: The loss of the space shuttle Columbia with seven astronauts on board
on Saturday was the most serious incident in NASA's shuttle program
since the 1986 explosion of the Challenger, in which seven astronauts died.

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1986, Jan 28 -- The shuttle Challenger explodes 72 seconds after
take-off, killing the seven crew, including the schoolteacher
Sharon C. McAuliffe. Blast caused by a leak of liquid hydrogen.
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1989, May 4-8 -- The shuttle Atlantis successfully releases payload, the Magellan spacecraft, for radar mapping of Venus.

1990, April 24-29 -- Discovery deploys the Hubble Space Telescope (news - web sites) into orbit.

1995 June 27-Jul 7 -- Atlantis marks the 100th U.S. human space launch with first docking between Shuttle and Russian Mir space station (news - web sites), f
errying Russian cosmonauts to and from the station. Dockings with Mir continue through 1996, 1997 until February 1998.

1998 Dec 4-15 -- Endeavour makes the first International Space Station (news - web sites) (ISS) flight to attach the first two modules of the ISS. ISS assemb
ly flights and crew exchange flights continue through 1999, 2000 and 2001.

2002 Nov-Dec -- Endeavour exchanges the ISS's fifth and sixth mission crews in a 14-day mission in orbit. It marks Endeavour's 19th flight.

2003 Feb 1 -- Columbia, the 113th shuttle mission, carrying Ilan Ramon, the first Israeli in space, breaks up over Texas with seven astronauts on board after
 16 days in orbit.


plain text stored in:text1.dd
[w418bxp@login01 Project 2]$
```

Figure 2.6 – Decryption of encrypted files

The below Figure 2.7 shows with the *ls* command, the decrypted files (with .ee suffix), denoting that the decryption was successful.

```
[w418bxp@login01 Project 2]$ ls
aes_decode_CBC_SP2026.cpp      aes_encode_COUNTER_SP2026.cpp  CEG47506750-Proj2-SP2026.doc  MSG1.e   MSG2.dd  MSG3.d   Task1CbcDec      text1     text1.ee
aes_decode_COUNTER_SP2026.cpp  aes_encode_ECB_SP2026.cpp      MSG1                          MSG1.ee  MSG2.e   MSG3.dd  Task1CbcEnc      text1.d
aes_decode_ECB_SP2026.cpp      ass2-input-files               MSG1.d                        MSG2     MSG2.ee  MSG3.e   Task2CounterDec  text1.dd
aes_encode_CBC_SP2026.cpp      ass2-input-files.zip           MSG1.dd                       MSG2.d   MSG3     MSG3.ee  Task2CounterEnc  text1.e
[w418bxp@login01 Project 2]$
```

Figure 2.7 – List of files

Task 2.1

So far, so good. For the subtask 2.1, we print the octal byte values using the command *od –b filename* for each file as shown in the below Figure 2.1.1.

```
w418bxp@login01:~/Project 2
[w418bxp@login01 Project 2]$ od -b MSG1.ee
0000000 146 346 223 010 031 271 056 063 356 104 312 214 107 360 177 276
0000020 111 011 333 232 051 347 265 016 024 347 314 001 115 371 271 126
0000040 377 204 110 272 042 325 213 206 020 336 212 322 062 041
0000056
[w418bxp@login01 Project 2]$ od -b MSG2.ee
0000000 146 346 223 010 031 271 056 063 356 104 312 214 107 360 177 276
0000020 111 001 326 232 051 347 265 016 024 347 314 001 115 371 271 126
0000040 377 204 110 272 042 325 213 206 020 336 212 322 062 041
0000056
[w418bxp@login01 Project 2]$ od -b MSG3.ee
0000000 146 346 223 010 031 271 056 063 356 104 312 214 107 360 177 276
0000020 111 011 333 232 051 347 265 016 024 347 314 001 115 371 271 126
0000040 377 204 110 272 042 325 213 206 020 336 212 321 062 041
0000056
[w418bxp@login01 Project 2]$
```

Figure 2.1.1 – Octal byte values of encrypted files

Task 2.2

From the above Figure 2.1.1, we can conclude for the comparison of MSG1.ee and MSG2.ee, the first 16 bytes (first row) are completely identical. At offset 0000020, the second byte of the second row shows the first difference, where MSG1.ee has 333 and MSG2.ee has 326, with subsequent bytes continuing to differ throughout.

For the comparison of MSG1.ee and MSG3.ee, the first 32 bytes (first two rows) are completely identical, with the only difference occurring at the third-last byte of the third row (offset 0000040 + 12), where MSG1.ee has 322 and MSG3.ee has 321.

This behavior is characteristic of AES-CTR mode, where ciphertext differences localize precisely to positions where plaintext differs. CTR mode transforms AES into a stream cipher by XORing plaintext with a keystream generated from successive counter values, ensuring byte level isolation of changes. Here, MSG1.ee and MSG3.ee differ by only one byte, indicating nearly identical plaintexts, while the differences between MSG1.ee and MSG2.ee begin at byte 17 and propagate from there, reflecting multiple plaintext variations from that position onward. Unlike ECB or CBC where changes ripple across entire blocks, CTR mode isolates differences with exact byte-level precision.

Task 2.3

We are required to use *diff* command to show the difference of two files (the original file, and the decrypted file). The result is shown in the Figure 2.3.1 below:

```
[w418bxp@login01 Project 2]$ diff MSG1 MSG1.dd
[w418bxp@login01 Project 2]$ diff MSG2 MSG2.dd
[w418bxp@login01 Project 2]$ diff MSG3 MSG3.dd
[w418bxp@login01 Project 2]$ diff text1 text1.dd
[w418bxp@login01 Project 2]$
```

Figure 2.3.1 – Difference between files

We can observe from the above Figure 2.3.1, we can clearly conclude that the encryption and decryption works perfectly as there are no differences shown for all the comparisons made.

As we have noticed already, the CBC mode requires an initialization vector (IV) to encrypt and decrypt a file where the size of the IV is set to the AES's default length. i.e. 16 bytes typically. The source code for both the files are provided in below Figure 2.6 and Figure 2.7.

```cpp
// CEG 47506750 sample code

#include<iostream>
#include<fstream>
#include<sstream>
#include<string>
using namespace std;

#include"cryptopp/cryptlib.h"
#include"cryptopp/hex.h"
#include"cryptopp/filters.h"
#include"cryptopp/des.h"
#include"cryptopp/aes.h"
#include"cryptopp/modes.h"

using namespace CryptoPP;

string aes_encode(string & plain,byte key[], byte iv[])
{
    string cipher;
    try{
        CTR_Mode<AES>::Encryption enc;
        enc.SetKeyWithIV(key, AES::DEFAULT_KEYLENGTH, iv);
        StringSource(plain, true, new StreamTransformationFilter(enc, new StringSink(cipher)));
    }
    catch(const CryptoPP::Exception & e)
    {
    }
    return cipher;
}

int main(int argc, char * argv[])
{
    fstream file1;
    fstream file2;
    byte key[AES::DEFAULT_KEYLENGTH];
    byte iv[AES::BLOCKSIZE];

    if(argc!=4)
    {
        cout<<"usage:aes_encode infile outfile key"<<endl;
        return 0;
    }
    file1.open(argv[1],ios::in);
    file2.open(argv[2],ios::out);
    //reading
    stringstream buffer;
    buffer << file1.rdbuf();
    string plain(buffer.str());
    //cout<<"plain text:"<<plain<<endl;
    //get key
    memset(key,0,AES::DEFAULT_KEYLENGTH);
    for(int i=0;i<AES::DEFAULT_KEYLENGTH;i++)
    {
        if(argv[3][i]!='\0')
        {
            key[i]=(byte)argv[3][i];
        }
        else
        {
            break;
        }
    }
    memset(iv, 0, AES::BLOCKSIZE);
    //print key
    string encoded;
    encoded.clear();
    StringSource(key, sizeof(key), true, new HexEncoder( new StringSink(encoded)));
    cout << "key: " << encoded<< endl;
    //encode
    cout << "plain text: " << plain << endl;
    string cipher=aes_encode(plain,key, iv);
    file2<<cipher;
    cout<<"cipher text stored in:"<<argv[2]<<endl;

    file1.close();
    file2.close();
}
```

Figure 2.6 – Source code (aes_encode_Counter_SP2026.cpp)

```
// CEG 47506750 sample code

#include<iostream>
#include<fstream>
#include<sstream>
#include<string>
using namespace std;

#include"cryptopp/cryptlib.h"
#include"cryptopp/hex.h"
#include"cryptopp/filters.h"
#include"cryptopp/des.h"
#include"cryptopp/aes.h"
#include"cryptopp/modes.h"

using namespace CryptoPP;
string aes_decode(string & cipher,byte key[], byte iv[])
{
    string plain;
    try{
        CTR_Mode< AES >::Decryption dec;
        dec.SetKeyWithIV(key, AES::DEFAULT_KEYLENGTH, iv);
        StringSource s(cipher, true, new StreamTransformationFilter(dec, new StringSink(plain)));
    }
    catch(const CryptoPP::Exception& e){
    }
    return plain;
}
int main(int argc,char * argv[])
{
    fstream file1;
    fstream file2;
    byte key[AES::DEFAULT_KEYLENGTH];
    byte iv[AES::BLOCKSIZE];

    if(argc!=4)
    {
        cout<<"usage:aes_decode infile outfile key"<<endl;
        return 0;
    }
    file1.open(argv[1],ios::in);
    file2.open(argv[2],ios::out);
    //reading
    stringstream buffer;
    buffer << file1.rdbuf();
    string cipher(buffer.str());
    //get key
    memset(key,0,AES::DEFAULT_KEYLENGTH);
    for(int i=0;i<AES::DEFAULT_KEYLENGTH;i++)
    {
        if(argv[3][i]!='\0')
        {
            key[i]=(byte)argv[3][i];
        }
        else
        {
            break;
        }
    }
    memset(iv, 0, AES::BLOCKSIZE);
    //print key
    string encoded;
    encoded.clear();
    StringSource(key, sizeof(key), true, new HexEncoder( new StringSink(encoded)));
    cout << "key: " << encoded<< endl;
    //decode
    string plain=aes_decode(cipher,key, iv);
    cout << "recovered text: " << plain<< endl;
    file2<<plain;
    cout<<"plain text stored in:"<<argv[2]<<endl;
}
```

Figure 2.7 – Source code (aes_decode_Counter_SP2026.cpp)